Daniel Zhao

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Education

University of Waterloo — **GPA: 4.0/4.0**
Honours Bachelor of Computer Science (Co-op) — *Sep 2024 – Present*

Technical Skills

Languages: C, C++, C#, Python, Java, Ruby, HTML, CSS, JavaScript, TypeScript, SQL, Racket
Technologies: React, React Native, Rails, Next.js, Express.js, Node.js, Flask, Django, Tailwind CSS, Unity
Tools: Git, Bash, Linux, Docker, Kubernetes, Figma, Postman, MongoDB, PostgreSQL, Firebase, AWS, GCP

Experience

Cineplex Digital Media — Sep 2025 – Dec 2025
Software Developer — *Waterloo, ON*

- Built an interactive analytics dashboard in **React**, **TypeScript** and **Tailwind CSS**, correlating **1,000+** daily Kinect-tracked physical movements with ad exposure insights to evaluate digital signage performance.
- Created a **Node.js** and **Express.js** backend with a **PostgreSQL** database, implementing **RESTful API** endpoints and data processing logic to align movement with ad playback events for fast dashboard queries.
- Developed a media analysis pipeline with **Python**, integrating **Gemini API** and **SAM 3** to automate content tagging, saving **200+ hours** of manual annotation work across hundreds of video and image files.
- Implemented a robust server-upload service in **C#** and resolved key **SQLite** data handling issues for a user analytics system deployed to over **70** digital signage units across **12** malls.

Waterloo Aerial Robotics Group — May 2025 – Aug 2025
Autonomy Software Developer — *Waterloo, ON*

- Developed an autonomous landing module in **Python**, implementing configurable target selection algorithms that enable drones to rank potential landing zones and land on targets with under **2cm** precision.
- Implemented real-time landing pad detection using **YOLOv8** and **OpenCV**, achieving **95%** detection reliability.
- Created unit and integration tests using **Pytest**, reinforcing maintainability and development best practices.

Waterloo Reality Labs — Sep 2024 – Apr 2025
Software Developer — *Waterloo, ON*

- Designed a context-aware prompting interface in **Unity** using **C#** and the **Meta XR SDK** to capture user gaze, proximity, and object interactions, in order to drive personalized feedback in a VR environment.
- Built a unified integration layer for **4 LLM backends**, supporting both cloud APIs and **Ollama** local inference.
- Integrated voice interaction with LLMs by implementing **OpenAI Whisper**, reducing user input time by **5x**.

Projects

Notation – AI LaTeX Transcription — *React, TypeScript, Rails, Gemini API, Docker, GCP*

- Built a **React** app that converts handwritten math PDFs and images into LaTeX using **Gemini API**.
- Implemented a **Ruby on Rails** backend to handle file processing and LaTeX compilation for text outputs.
- Utilized **Docker** for containerized deployment on **GCP**, with **CI/CD** for automated builds and deploys.

ChordCraft – CRUD Music Notes — *React, JavaScript, Node.js, MongoDB, ABCJS*

- Developed a **React** app with an interactive piano interface that renders sheet music in real time using **ABCJS**
- Built a **Node.js** and **Express.js** backend with **MongoDB** for user authentication and persistent storage.
- Deployed the app and supported **100+** unique users creating, editing, and saving musical compositions.

Terra – AR World Building — *C#, Unity, AR Foundation, ARCore*

- Created an AR mobile app that allows users to place 3D blocks in physical spaces using **Unity** and **ARCore**.
- Developed **C#** scripts to implement object placement and color selection for block customization.
- Implemented raycasting and intuitive touch controls, improving object placement precision by **90%**.

MotionWave – Vision AI Harmonization — *Next.js, TypeScript, MediaPipe, Web Audio API*

- Developed a gesture-controlled music app with **Next.js**, using **MediaPipe** for real-time hand tracking.
- Integrated a **WebAssembly** neural network for SATB harmony generation, maintaining **60FPS** UI performance.
- Demonstrated live to **50+** people, communicating technical design and validating UX through audience testing.